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Saveetha Institute of Medical And Technical Sciences
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SAVEETHA SCHOOL OF ENGINEERING, SIMATS
Department of Computer Science and Engineering
Assignment Information

Particular	Details
Name:	Dinnepati Sindhu Prasad
Register Number:	192311271
Department:	BE
Programme:	CSE
Course Code & Course Name	CSA1522 – Cloud Computing and Big Data Analytics
Academic Year / Batch	2023–27
Faculty Name	Dr P Rajaram
Assignment Title	Cloud-Based Placement Data Management and Big Data Analytics using JOHO
Date of Issue	31-08-2026
Date of Submission	01-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Understand the cloud services with different service models. CO2: Demonstrate the deployment of Virtualization and build a data center using virtualization tools. CO3: Build methods to access cloud and various techniques to collaborate within cloud.
Bloom's Taxonomy Level	L3 – Apply and L4 – Analyse
SDG Mapping	SDG 4 – Quality Education SDG 8 – Decent Work and Economic Growth SDG 9 – Industry Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities
Industry	Applies cloud services, virtualization and distributed big-data processing to a JOHO-based placement workflow, supporting scalable access, collaboration and analytics while considering security, privacy and responsible data use.

Problem Statement:

Design and evaluate a simple cloud-based placement data management and access solution for a university using the JOHO platform. The placement cell needs a reliable way to organize placement information and provide controlled access to students and placement staff. Students shall select suitable IaaS, PaaS and SaaS service models, design a basic cloud architecture, create and configure virtual machines using Oracle VM VirtualBox, demonstrate secure access to a cloud-hosted web application or Web API, and show how users can collaborate through shared cloud resources.

Assessment Rubrics

Assessment Criterion / Marks	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
CO1 – Cloud Service Models & Architecture 25 Marks	Correctly identifies the JOHO requirements, selects appropriate IaaS/PaaS/SaaS models, and presents a complete architecture with clear justification.	Selects appropriate service models and presents a suitable architecture with minor gaps in justification or detail.	Identifies basic service models and provides a simple architecture, but the justification or mapping to requirements is limited.	Service models are unsuitable or misunderstood; architecture is missing/unclear and justification is inadequate.
CO2 – Virtualization & VM Deployment 25 Marks	Creates and configures two VMs correctly using Oracle VM VirtualBox; records CPU, memory, storage and networking; demonstrates communication and clearly explains the hypervisor.	Creates and configures both VMs successfully with mostly correct resources and networking; communication works with minor issues.	Creates the VMs and demonstrates basic operation, but configuration, networking, testing or explanation is incomplete.	VM creation/configuration is incorrect or incomplete; communication is not demonstrated and supporting evidence is insufficient.
CO3 – Cloud Access & Collaboration 25 Marks	Demonstrates secure cloud access successfully, including authentication/access control, clear request-response evidence and a practical collaboration workflow.	Demonstrates working cloud access and collaboration with minor gaps in security, explanation or evidence.	Demonstrates basic cloud access, but collaboration, authentication or security evidence is limited.	Cloud access or collaboration is non-functional, unclear, insecure or not supported by evidence.
Analysis, Validation & Engineering Justification 15 Marks	Compares feasible alternatives using clear criteria, performs relevant tests, interprets results accurately and gives a well-supported final decision.	Provides a reasonable comparison and testing with generally correct interpretation and justification.	Provides limited comparison or testing; interpretation and justification are basic.	Little or no meaningful comparison, testing, interpretation or engineering justification is provided.
Security, Privacy, Ethics & SDG Relevance 10 Marks	Clearly addresses security, privacy, ethical handling of placement data and relevant SDGs with specific application to the JOHO scenario.	Addresses relevant security, privacy, ethical and SDG considerations with minor omissions.	Mentions general security, privacy, ethics or SDG points but gives limited connection to the problem.	Important security, privacy, ethical or SDG considerations are missing or inappropriate.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
3. Github and GCR Upload

1. Problem Statement

The university placement cell needs a reliable cloud-based system to manage placement information and provide controlled access to students and placement staff. The solution uses the **Zoho Creator platform** for placement data management, secure access and collaboration. Virtual machines are created and configured using **VMware Workstation Pro** to demonstrate virtualization and network communication between a server and client.

2. Objective

- To develop a simple cloud-based placement data management system.
- To organize student, company, placement drive and placement result information.
- To provide controlled and secure access to users.
- To understand and apply IaaS, PaaS and SaaS cloud service models.
- To create and configure two virtual machines.
- To demonstrate communication between the virtual machines.
- To enable users to collaborate using shared cloud resources.

3. Requirements and Environment Used

Software Requirements

- **Zoho Creator** – Cloud application development and data management.
- **VMware Workstation Pro** – Virtual machine creation and configuration.
- **Ubuntu Linux** – Operating system for virtual machines.
- **Web Browser** – Accessing Zoho Creator.

VM Configuration

Resource	Placement-Server	Placement-Client
RAM	2 GB	2 GB
CPU	2 Processors	2 Processors
Storage	20 GB	20 GB
Network	Host-only	Host-only

Constraints

- Simple and low-cost implementation.
- Limited VM resources.
- Secure access to placement information.
- Reliable communication between virtual machines.

4. Application of Cloud Concepts

4.1 IaaS

Infrastructure as a Service (IaaS) provides virtual computing resources such as CPU, memory, storage and networking. The two Ubuntu virtual machines created using VMware demonstrate the concept of virtual infrastructure.

4.2 PaaS

Platform as a Service (PaaS) provides a platform for developing applications without managing the underlying infrastructure. **Zoho Creator** is used as the application development platform.

4.3 SaaS

Software as a Service (SaaS) provides software through the internet. The completed placement management application is accessed through the Zoho Creator cloud platform.

4.4 Virtualization

Virtualization allows multiple virtual computers to run on a single physical computer. Two VMs, **Placement-Server** and **Placement-Client**, were created using VMware Workstation Pro.

4.5 Security and Access Control

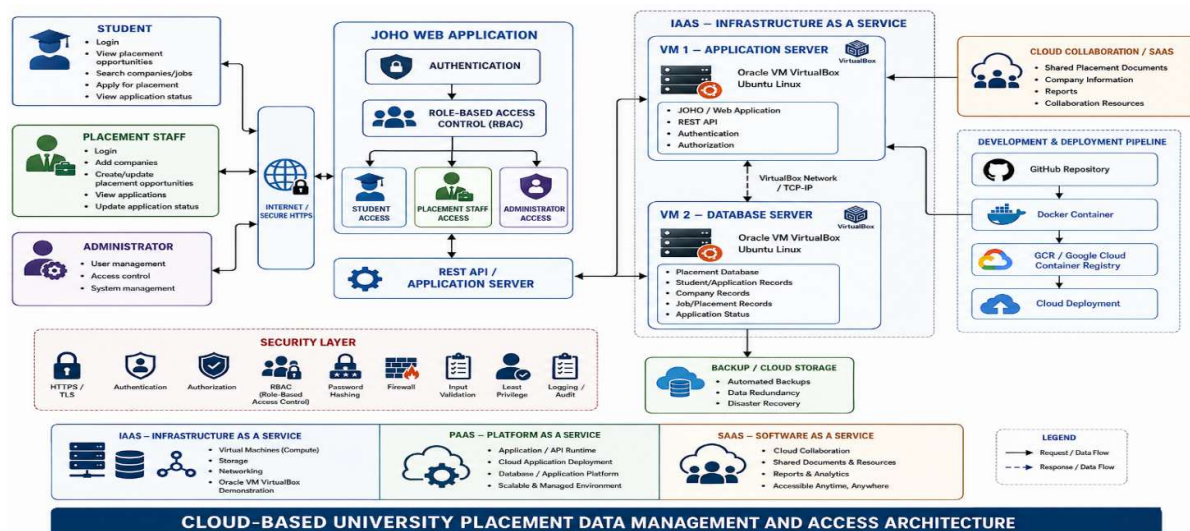
The Zoho Creator application provides controlled access to placement information. Different users such as students and placement staff can access the required information according to their roles.

4.6 Cloud Collaboration

The placement application provides a shared cloud environment where placement staff can manage placement information and students can access relevant placement details.

5. Proposed Solution / Cloud Architecture

The proposed system combines Zoho Creator and VMware virtualization.



Zoho Creator manages the cloud-based placement application, while VMware provides the virtualized server and client environment.

6. Implementation

6.1 Zoho Creator – Placement Management

A placement management application was created using Zoho Creator. It contains modules for managing:

- Student Profiles
- Company Profiles
- Placement Drives
- Placement Results
- Placement Reports
- Dashboard and Analytics

The system provides a centralized location for managing placement information.

6.2 Zoho Creator – Secure Access & Collaboration

The application provides controlled access to placement information for students and placement staff. Placement staff can update placement information, while students can view relevant placement details. This allows users to collaborate using shared cloud resources.

6.3 VMware – Virtual Machine Configuration

Two Ubuntu virtual machines were created:

- **Placement-Server**
- **Placement-Client**

Both VMs were configured with 2 GB RAM, 2 processors and 20 GB storage.

6.4 VMware – Network Configuration

Both VMs were configured using a **Host-only network**.

The IP addresses were:

- Placement-Server: **192.168.204.128**
- Placement-Client: **192.168.204.129**

Communication was tested using the ping command.

7. Pseudo Code

START

1. Open the Zoho Creator Placement Management application.
2. Add student details.
3. Add company details.
4. Create placement drive details.
5. Store and manage placement results.

6. Provide controlled access to students and placement staff.
 7. Allow users to access and share placement information through the cloud.
 8. Create Placement-Server virtual machine in VMware.
 9. Create Placement-Client virtual machine in VMware.
 10. Configure CPU, RAM, storage and network for both VMs.
 11. Assign Host-only network to both VMs.
 12. Obtain the IP address of both VMs.
 13. Test Client → Server communication using ping.
 14. Test Server → Client communication using ping.
 15. If communication is successful,
 Record the result.
 Else,
 Check the network configuration.
 16. Verify that all placement and virtualization requirements are satisfied.
- END

8. Test Cases and Results

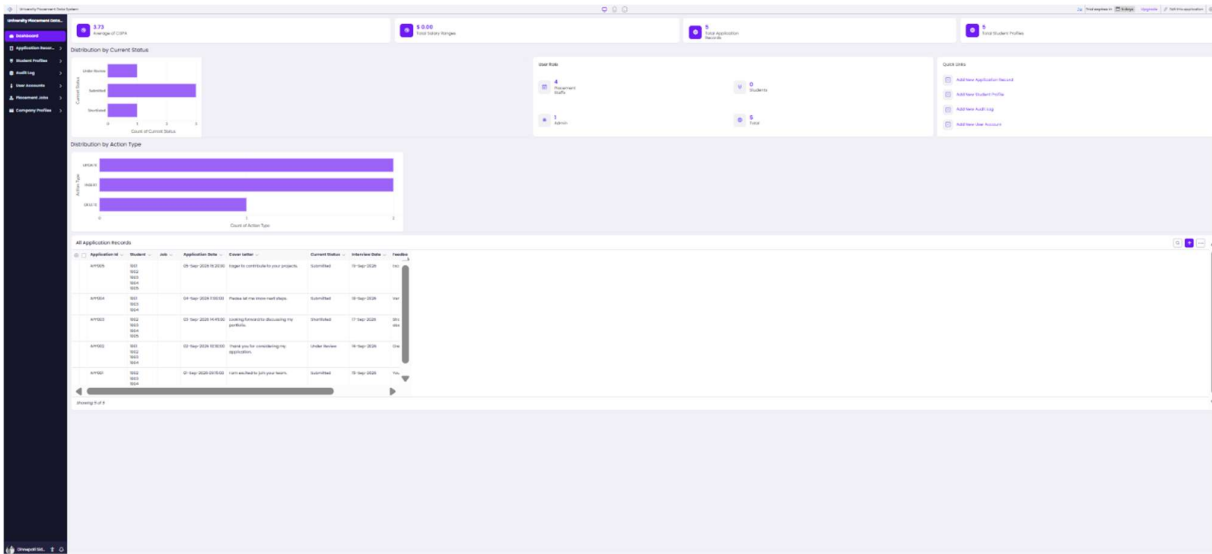
Test Case	Expected Result	Actual Result	Status
Add student details	Details should be stored	Details stored	Pass
Add company details	Company should be stored	Company stored	Pass
Create placement drive	Drive should be created	Drive created	Pass
View placement information	Information should be displayed	Information displayed	Pass
Client → Server communication	Successful communication	4/4 packets received	Pass
Server → Client communication	Successful communication	4/4 packets received	Pass

Network Test Result: 4 packets transmitted, 4 received, **0% packet loss.**

9. Execution Screenshots / Outputs

The following screenshots are included as evidence:

1. Zoho Creator placement dashboard.



2. Student Profile module.

The screenshot displays the Student Profile module. It features a sidebar with navigation options: Dashboard, Application Records, Student Profiles, Audit Log, User Accounts, Placement Jobs, and Company Profiles. The main content area shows a table titled 'All Student Profiles' with columns for Student ID, User, Enrollment Number, Branch, CGPA, Graduation Year, Resume URL, and Is Eligible. The table lists several student profiles with their respective details.

Student ID	User	Enrollment Number	Branch	CGPA	Graduation Year	Resume URL	Is Eligible
1005	U001 U002	EN2026005	Chemical Engineering	3.55	2030	https://example.com/resume5.pdf	true
1004	U001 U002 U003 U004 U005	EN2026004	Electrical Engineering	3.65	2030	https://example.com/resume4.pdf	true
1003	U001 U002 U003 U004 U005	EN2026003	Civil Engineering	3.90	2030	https://example.com/resume3.pdf	false
1002	U001 U002 U003 U004	EN2026002	Mechanical Engineering	3.72	2030	https://example.com/resume2.pdf	true
1001	U001 U002 U003 U005	EN2026001	Computer Science	3.85	2030	https://example.com/resume1.pdf	true

3. Company Profile module.

University Placement Data System

26 Trial expires in 5 days Upgrade Edit this application Help

University Placement Data... All Company Profiles

Company Code	Company Name	Industry	Website	Contact Email	Contact Phone	Office Address	Is Verified
					+9134081959		false
					+9072578582		false
					+7023543431		false

Showing 3 of 3

4. Placement Drive module.

University Placement Data System

26 Trial expires in 5 days Upgrade Edit this application Help

University Placement Data... All Placement Jobs

Job Code	Company	Job Title	Job Description	Required Skills	Salary Range	Application Deadline	Is Active
							false
							false
							false

Showing 3 of 3

Show Summary

5. Placement Result module.

University Placement Data System

26 Trial expires in 5 days Upgrade Edit this application Help

University Placement Data System

Placement Job

Job Code

Company

Job Title

Job Description

Required Skills

Salary Range

Application Deadline

☐ Is Active

6. VMware Placement-Server configuration.

Virtual Machine Settings

Hardware Options

Device	Summary
Memory	2 GB
Processors	2
Hard Disk (SCSI)	20 GB
Hard Disk 2 (SCSI)	1 GB
CD/DVD (SATA)	Auto detect
Network Adapter	Host-only
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: MB

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -

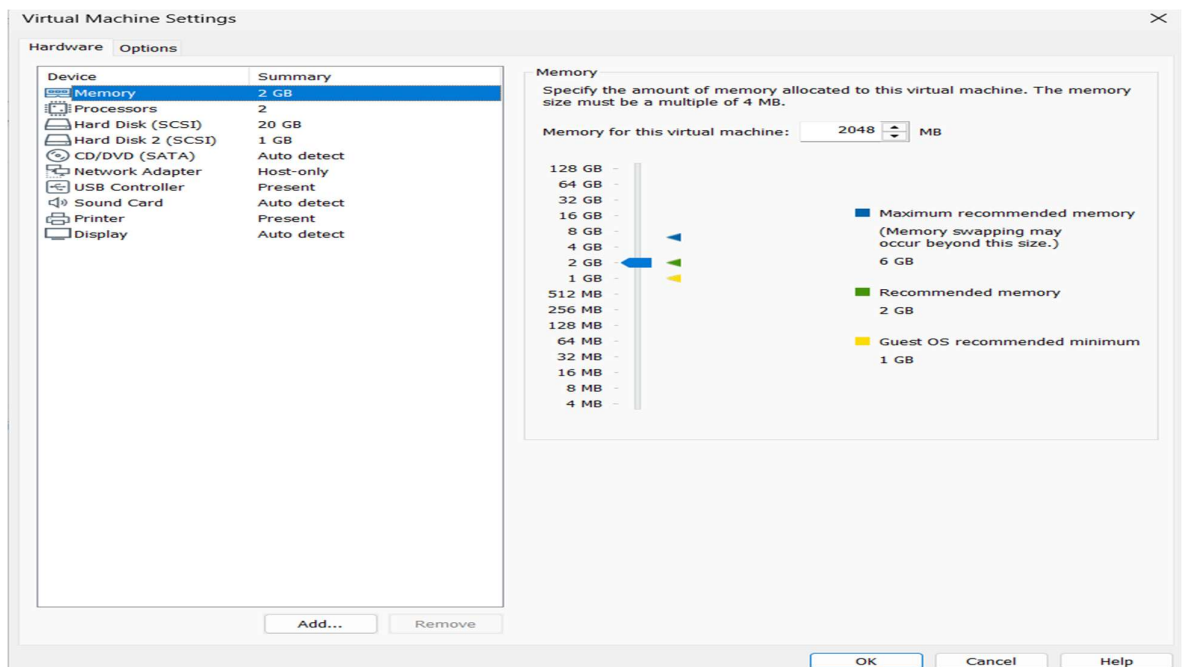
512 MB -
256 MB -
128 MB -
64 MB -
32 MB -
16 MB -
8 MB -
4 MB -

Maximum recommended memory
(Memory swapping may occur beyond this size.)
6 GB

Recommended memory
2 GB

Guest OS recommended minimum
1 GB

7. VMware Placement-Client configuration.



8. Placement-Server IP address.

```
File Edit View Search Terminal Help
    valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:7d:3c:87 brd ff:ff:ff:ff:ff:ff
    inet 192.168.204.128/24 brd 192.168.204.255 scope global dynamic noprefixrou
te ens33
    valid_lft 1747sec preferred_lft 1747sec
    inet6 fe80::666c:9e7e:7e33:d290/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

9. Placement-Client IP address.

```
File Edit View Search Terminal Help
dhilip@Placement-Client:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5c:07:f3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.204.129/24 brd 192.168.204.255 scope global dynamic noprefixroute ens33
        valid_lft 1732sec preferred_lft 1732sec
    inet6 fe80::4629:c605:110a:572a/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

10.

11. Client-to-Server successful ping.

```

PING 192.168.204.129 (192.168.204.129) 56(84) bytes of data.
64 bytes from 192.168.204.129: icmp_seq=1 ttl=64 time=4.13 ms
64 bytes from 192.168.204.129: icmp_seq=2 ttl=64 time=1.71 ms
64 bytes from 192.168.204.129: icmp_seq=3 ttl=64 time=3.06 ms
64 bytes from 192.168.204.129: icmp_seq=4 ttl=64 time=1.61 ms

--- 192.168.204.129 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 1.612/2.630/4.134/1.040 ms

```

12. Server-to-Client successful ping.

```

PING 192.168.204.128 (192.168.204.128) 56(84) bytes of data.
64 bytes from 192.168.204.128: icmp_seq=1 ttl=64 time=4.31 ms
64 bytes from 192.168.204.128: icmp_seq=2 ttl=64 time=1.81 ms
64 bytes from 192.168.204.128: icmp_seq=3 ttl=64 time=2.24 ms
64 bytes from 192.168.204.128: icmp_seq=4 ttl=64 time=1.08 ms

--- 192.168.204.128 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.085/2.362/4.312/1.199 ms

```

10. Analysis, Trade-offs and Justification

Zoho Creator was selected because it provides a simple cloud platform for developing and managing applications without requiring a separate application server. VMware Workstation Pro was used to demonstrate virtualization using two VMs.

The solution is simple and suitable for a university placement cell. However, the VM resources are limited and the system is intended mainly for demonstration rather than large-scale production use.

11. Broader Considerations / SDG Relevance

The system supports **SDG 4 – Quality Education** by improving access to organized placement information and supporting digital services for students.

Cloud-based data management also reduces the need for maintaining multiple physical systems and supports efficient resource utilization.

12. Conclusion, Limitations and Future Improvements

The cloud-based placement management solution was successfully developed using Zoho Creator. Placement information can be organized and accessed through the cloud. Two virtual machines were created using VMware Workstation Pro and successfully communicated with each other with **0% packet loss**.

Limitations

- Designed mainly for academic demonstration.
- Limited VM resources.
- Basic access-control implementation.

Future Improvements

- Add email/SMS notifications.
- Add advanced authentication.
- Add placement prediction and analytics.
- Deploy a dedicated production Web API.

12. Individual Contribution

K. Dhillip – Zoho Creator application development, placement modules, cloud concepts, VMware virtual machine configuration, network configuration and testing.

13. One-Page Individual Reflection

Through this assignment, I gained practical knowledge of cloud computing and virtualization. I learned how different cloud service models such as IaaS, PaaS and SaaS can be applied to a real-world university placement system.

I used **Zoho Creator** to develop a cloud-based placement management application and learned how cloud platforms can be used to organize and share information. I also learned the importance of controlled access and collaboration when multiple users work with shared data.

I created two virtual machines, **Placement-Server** and **Placement-Client**, using VMware Workstation Pro. I configured their CPU, RAM, storage and network settings. I also learned how to configure a Host-only network and verify communication using the ping command. Both VMs successfully communicated with each other with 0% packet loss.

One challenge was initially getting communication between the two VMs while using NAT networking. I solved this by configuring both VMs to use a Host-only network. This helped me understand practical virtual networking rather than only learning the concept theoretically.

This assignment improved my understanding of cloud platforms, virtualization, networking, access control and collaborative cloud applications. In the future, I would improve the system by adding stronger authentication, notifications and advanced placement analytics.

The project is relevant to **SDG 4 – Quality Education**, as it supports digital access to organized placement information and improves the efficiency of university placement activities.

14. References

1. **Zoho Creator** – Cloud application development platform.
2. **VMware Workstation Pro** – Virtualization platform.
3. **Ubuntu Linux** – Operating system used for virtual machines.
4. Course notes and classroom materials on **Cloud Computing and Virtualization**.

